

Appendix: material indices

Introduction and synopsis

The performance, P , of a component is characterized by a performance equation. The performance equation contains groups of material properties. These groups are the material indices. Sometimes the "group" is a single property; thus if the performance of a beam is measured by its stiffness, the performance equation contains only one property, the elastic modulus E . It is the material index for this problem. More commonly the performance equation contains a group of two or more properties. Familiar examples are the specific stiffness, E/ρ , and the specific strength, σ_y/ρ , (where σ_y is the yield strength or elastic limit, and ρ is the density), but there are many others. They are a key to the optimal selection of materials. Details of the method, with numerous examples are given in Chapters 5 and 6 and in the book "Case studies in materials selection". This Appendix compiles indices for a range of common applications.

Uses of material indices

Material selection. Components have functions: to carry loads safely, to transmit heat, to store energy, to insulate, and so forth. Each function has an associated material index. Materials with high values of the appropriate index maximize that aspect of the performance of the component. For reasons given in Chapter 5, the material index is generally independent of the details of the design. Thus the indices for beams in the tables that follow are independent of the detailed shape of the beam; that for minimizing thermal distortion of precision instruments is independent of the configuration of the instrument, and so forth. This gives them great generality.

Material deployment or substitution. A new material will have potential application in functions for which its indices have unusually high values. Fruitful applications for a new material can be identified by evaluating its indices and comparing them with those of existing, established materials. Similar reasoning points the way to identifying viable substitutes for an incumbent material in an established application.

How to read the tables. The indices listed in the Tables 1 to 7 are, for the most part, based on the objective of minimizing mass. To minimize cost, use the index for minimum mass, replacing the density ρ by the cost per unit volume, $C_m\rho$, where C_m is the cost per kg. To minimize energy content or CO₂ burden, replace ρ by $H_p\rho$ or by $CO_2\rho$ where H_p is the production energy per kg and CO_2 is the CO₂ burden per kg.

Table A1 Stiffness-limited design at minimum mass (cost, energy, eco-impact)

FUNCTION and CONSTRAINTS	Maximize
TIE (tensile strut) stiffness, length specified; section area free	E / ρ
SHAFT (loaded in torsion) stiffness, length, shape specified, section area free stiffness, length, outer radius specified; wall thickness free stiffness, length, wall-thickness specified; outer radius free	$G^{1/2} / \rho$ G / ρ $G^{1/3} / \rho$
BEAM (loaded in bending) stiffness, length, shape specified; section area free stiffness, length, height specified; width free stiffness, length, width specified; height free	$E^{1/2} / \rho$ E / ρ $E^{1/3} / \rho$
COLUMN (compression strut, failure by elastic buckling) buckling load, length, shape specified; section area free	$E^{1/2} / \rho$
PANEL (flat plate, loaded in bending) stiffness, length, width specified, thickness free	$E^{1/3} / \rho$
PLATE (flat plate, compressed in-plane, buckling failure) collapse load, length and width specified, thickness free	$E^{1/3} / \rho$
CYLINDER WITH INTERNAL PRESSURE elastic distortion, pressure and radius specified; wall thickness free	E / ρ
SPHERICAL SHELL WITH INTERNAL PRESSURE elastic distortion, pressure and radius specified, wall thickness free	$E / (1 - \nu) \rho$

Table A2 Strength-limited design at minimum mass (cost, energy, eco-impact)

FUNCTION and CONSTRAINTS	Maximize
TIE (tensile strut) stiffness, length specified; section area free	σ_f / ρ
SHAFT (loaded in torsion) load, length, shape specified, section area free load, length, outer radius specified; wall thickness free load, length, wall-thickness specified; outer radius free	$\sigma_f^{2/3} / \rho$ σ_f / ρ $\sigma_f^{1/2} / \rho$
BEAM (loaded in bending) load, length, shape specified; section area free load length, height specified; width free load, length, width specified; height free	$\sigma_f^{2/3} / \rho$ σ_f / ρ $\sigma_f^{1/2} / \rho$
COLUMN (compression strut) load, length, shape specified; section area free	σ_f / ρ
PANEL (flat plate, loaded in bending) stiffness, length, width specified, thickness free	$\sigma_f^{1/2} / \rho$
PLATE (flat plate, compressed in-plane, buckling failure) collapse load, length and width specified, thickness free	$\sigma_f^{1/2} / \rho$
CYLINDER WITH INTERNAL PRESSURE elastic distortion, pressure and radius specified; wall thickness free	σ_f / ρ
SPHERICAL SHELL WITH INTERNAL PRESSURE elastic distortion, pressure and radius specified, wall thickness free	σ_f / ρ
FLYWHEELS, ROTATING DISKS maximum energy storage per unit volume; given velocity maximum energy storage per unit mass; no failure	ρ σ_f / ρ

Table A3 Strength-limited design: springs, hinges etc for maximum performance

FUNCTION and CONSTRAINTS	Maximize
SPRINGS maximum stored elastic energy per unit volume; no failure maximum stored elastic energy per unit mass; no failure	σ_f^2 / E $\sigma_f^2 / E \rho$
ELASTIC HINGES radius of bend to be minimized (max flexibility without failure)	σ_f / E
KNIFE EDGES, PIVOTS minimum contact area, maximum bearing load	σ_f^3 / E^2 and H
COMPRESSION SEALS AND GASKETS maximum conformability; limit on contact pressure	$\sigma_f^{3/2} / E$ and $1/E$
DIAPHRAGMS maximum deflection under specified pressure or force	$\sigma_f^{3/2} / E$
ROTATING DRUMS AND CENTRIFUGES maximum angular velocity; radius fixed; wall thickness free	σ_f / ρ

Table A4 Vibration-limited design

FUNCTION and CONSTRAINTS	Maximize
TIES, COLUMNS maximum longitudinal vibration frequencies	$\sqrt{E} / \rho$
BEAMS, all dimensions prescribed maximum flexural vibration frequencies BEAMS, length and stiffness prescribed maximum flexural vibration frequencies	$\sqrt{E} / \rho$ $E^{1/2} / \rho$
PANELS, all dimensions prescribed maximum flexural vibration frequencies PANELS, length, width and stiffness prescribed maximum flexural vibration frequencies	E / ρ $E^{1/3} / \rho$
TIES, COLUMNS, BEAMS, PANELS, stiffness prescribed minimum longitudinal excitation from external drivers, ties minimum flexural excitation from external drivers, beams minimum flexural excitation from external drivers, panels	$\eta E / \rho$ $\eta E^{1/2} / \rho$ $\eta E^{1/3} / \rho$

Table A5 Damage-tolerant design

FUNCTION and CONSTRAINTS	Maximize
TIES (tensile member) maximum flaw tolerance and strength, load-controlled design maximum flaw tolerance and strength, displacement-control maximum flaw tolerance and strength, energy-control	K_{Ic} and σ_f K_{Ic} / E and σ_f K_{Ic}^2 / E and σ_f
SHAFTS (loaded in torsion) maximum flaw tolerance and strength, load-controlled design maximum flaw tolerance and strength, displacement-control maximum flaw tolerance and strength, energy-control	K_{Ic} and σ_f K_{Ic} / E and σ_f K_{Ic}^2 / E and σ_f
BEAMS (loaded in bending) maximum flaw tolerance and strength, load-controlled design maximum flaw tolerance and strength, displacement-control maximum flaw tolerance and strength, energy-control	K_{Ic} and σ_f K_{Ic} / E and σ_f K_{Ic}^2 / E and σ_f
PRESSURE VESSEL yield-before-break leak-before-break	K_{Ic} / σ_f K_{Ic}^2 / σ_f

Table A6 Thermal and thermo-mechanical design

FUNCTION and CONSTRAINTS	Maximize
THERMAL INSULATION MATERIALS minimum heat flux at steady state; thickness specified minimum temp rise in specified time; thickness specified minimize total energy consumed in thermal cycle (kilns, etc)	$1 / \lambda$ $1 / a = \rho C_p / \lambda$ $\sqrt{a} / \lambda = \sqrt{1 / \lambda \rho C_p}$
THERMAL STORAGE MATERIALS maximum energy stored / unit material cost (storage heaters) maximize energy stored for given temperature rise and time	C_p / C_m $\lambda / \sqrt{a} = \sqrt{\lambda \rho C_p}$
PRECISION DEVICES minimize thermal distortion for given heat flux	λ / a
THERMAL SHOCK RESISTANCE maximum change in surface temperature; no failure	$\sigma_f / E \alpha$
HEAT SINKS maximum heat flux per unit volume; expansion limited maximum heat flux per unit mass; expansion limited	$\lambda / \Delta \alpha$ $\lambda / \rho \Delta \alpha$
HEAT EXCHANGERS (pressure-limited) maximum heat flux per unit area; no failure under Δp maximum heat flux per unit mass; no failure under Δp	$\lambda \sigma_f$ $\lambda \sigma_f / \rho$

Table A7 Electro-mechanical design

FUNCTION and CONSTRAINTS	Maximize
BUS BARS minimum life-cost; high current conductor	$1 / \rho_e \rho C_m$
ELECTRO-MAGNET WINDINGS maximum short-pulse field; no mechanical failure maximize field and pulse-length, limit on temperature rise	σ_f $C_p \rho / \rho_e$
WINDINGS, HIGH-SPEED ELECTRIC MOTORS maximum rotational speed; no fatigue failure minimum ohmic losses; no fatigue failure	σ_e / ρ_e $1 / \rho_e$
RELAY ARMS minimum response time; no fatigue failure minimum ohmic losses; no fatigue failure	$\sigma_e / E \rho_e$ $\sigma_e^2 / E \rho_e$